

CSE331 Theory

Assignment 04

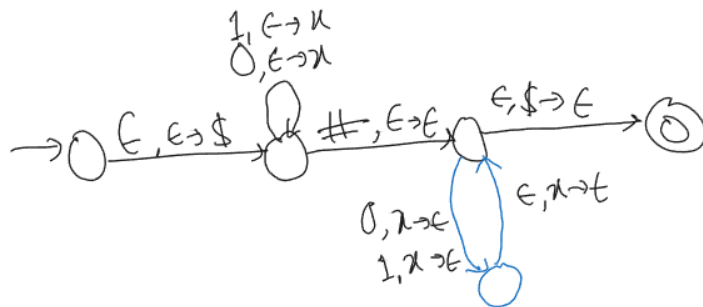
1. Draw PDA for the following languages

$$L_1 = \{ w \in \{0, 1\}^* \mid w \text{ ends with } 01 \}$$

$$L_2 = \{ w_1, w_2 \in \{0, 1\} \mid w_1 \# w_2 \text{ where } |w_1| = |w_2| \}$$

$$L_3 = \{ w_1, w_2 \in \{0, 1\} \mid w_1 \# w_2 \text{ where } |w_1| \neq |w_2| \}$$

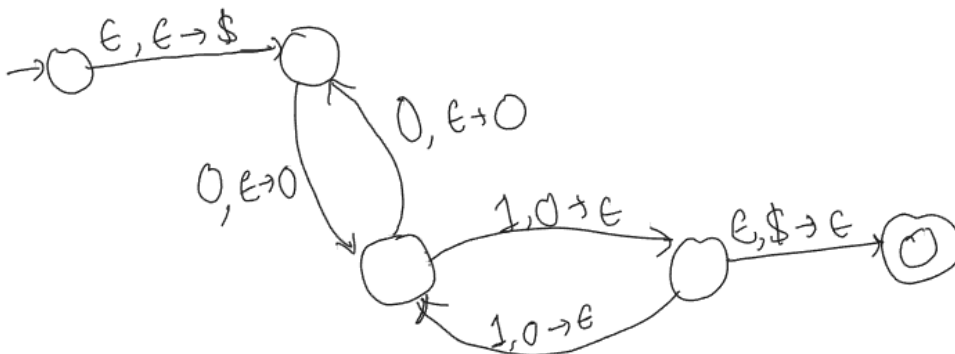
$$L_4 = \{ w_1, w_2 \in \{0, 1\} \mid w_1 \# w_2 \text{ where } |w_1| = 2|w_2| \}$$



$$L_5 = \{ w \in \{a, b\} \mid \text{the number of 'a' and 'b' are not equal} \}$$

Shown in class

$$L_6 = \{ w \in \{0, 1\} \mid 0^n 1^n \text{ where } n \text{ is odd} \}$$



$$L_7 = \{ w \in \{0, 1, 2\} \mid 0^i 1^j 2^k \text{ where } j=k \text{ \& } i, j, k \geq 1 \}$$

$$L_8 = \{ w \in \{0, 1, 2\} \mid 0^i 1^j 2^k \text{ where } i=j+k \text{ \& } i, j, k \geq 0 \}$$

$$L_9 = \{ w \in \{0, 1, 2\} \mid 0^i 1^j 2^k \text{ where } k = i+2j \text{ \& } i, j, k \geq 0 \}$$

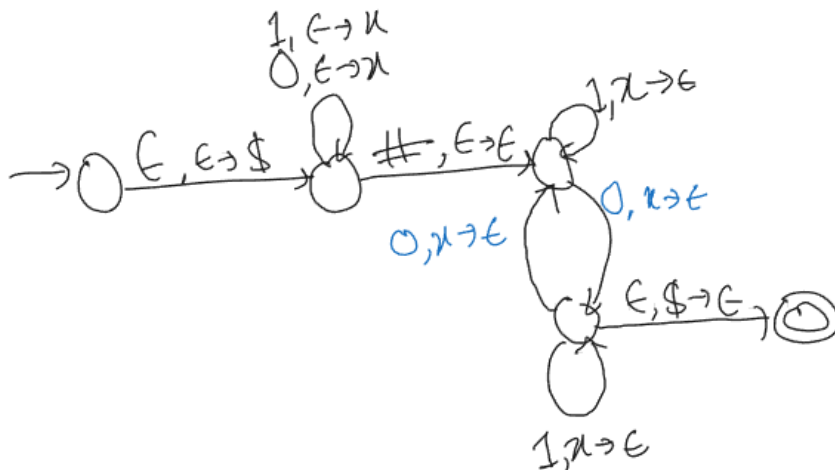
$$L_{10} = \{ w \in \{0, 1, 2\} \mid 0^i 1^j 2^k \text{ where } k > i+j \text{ \& } i, j, k \geq 0 \}$$

$$L_{11} = \{ w \in \{0, 1, 2\} \mid 0^i 1^j 2^k \text{ where } k > i+3j \text{ \& } i, j, k \geq 0 \}$$

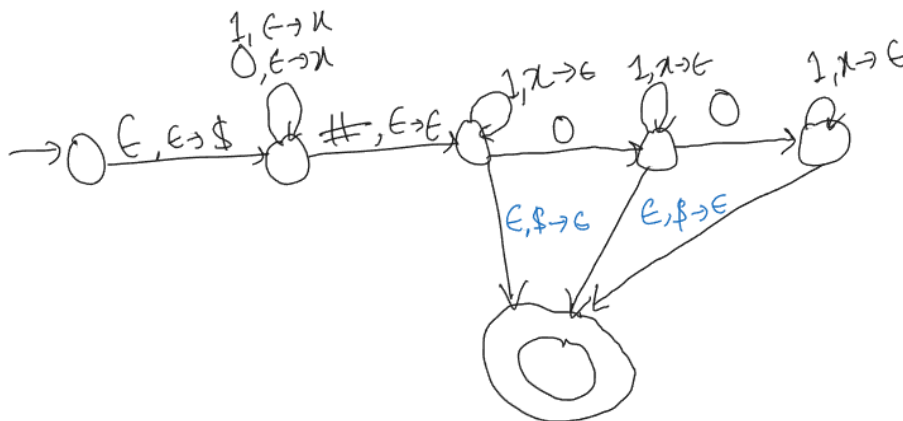
$$L_{12} = \{ w \in \{a, b\} \mid \text{number of 'a' is 2 more than number of 'b'} \}$$

$$L_{13} = \{ w \in \{0, 1\} \mid \text{every second letter is 1} \}$$

$$L_{14} = \{ w_1, w_2 \in \{0, 1\} \mid w_1 \# w_2 \text{ where } |w_1| = |w_2| \text{ and } w_2 \text{ has odd number of 0s} \}$$



$$L_{15} = \{ w_1, w_2 \in \{0, 1\} \mid w_1 \# w_2 \text{ where } |w_1| = |w_2| \text{ where } w_2 \text{ has at most two 0s} \}$$



$$L_{16} = \{ w_1, w_2 \in \{0, 1\} \mid w_1 \# w_2 \text{ where } |w_1| = |w_2| \text{ where } w_2 \text{ has exactly two 0s} \}$$

$$L_{17} = \{ w_1, w_2 \in \{0, 1\} \mid w_1 a w_2 \text{ where } \text{count}(w_1, 0) \text{ is same as } \text{count}(w_2, 1) \}$$

$$L_{18} = \{ w_1, w_2, w_3 \in \{0, 1\} \mid w_1 \# w_2 \# w_3 \text{ where } \text{count}(w_1, 0) \text{ is same as } \text{count}(w_2, 0) \text{ or } \text{count}(w_2, 1) \text{ is the same as } \text{count}(w_3, 1) \}$$

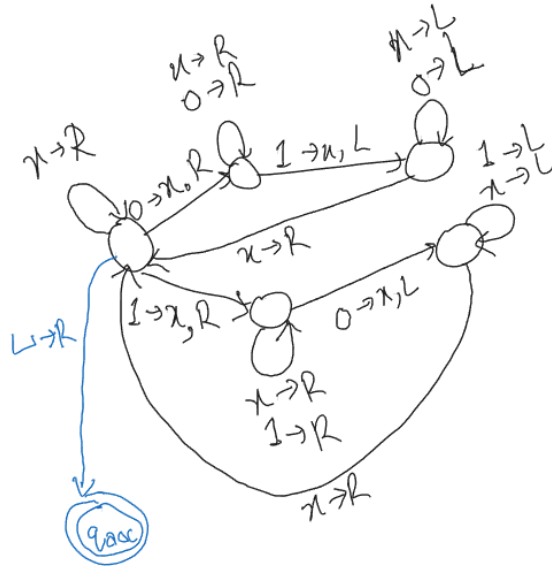
2. Draw Turning Machine for the following languages

$L_1 = \{ w \in \{0, 1\}^* \mid w \text{ is a palindrome} \}$

$L_2 = \{ w \in \{0, 1, 2\}^* \mid 0^n 1^n 2^n \text{ where } n \geq 0 \}$

Similar Example - [Link](#)

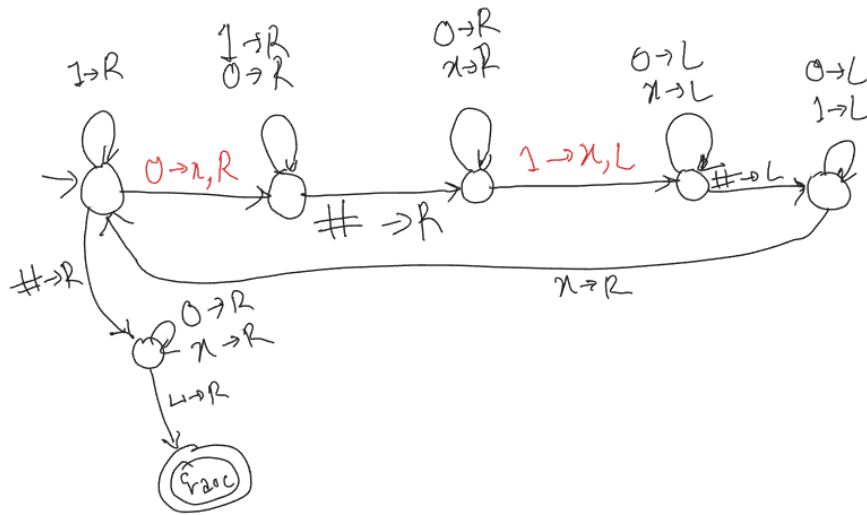
$L_3 = \{ w \in \{0, 1\}^* \mid \text{equal number of 0 and 1} \}$



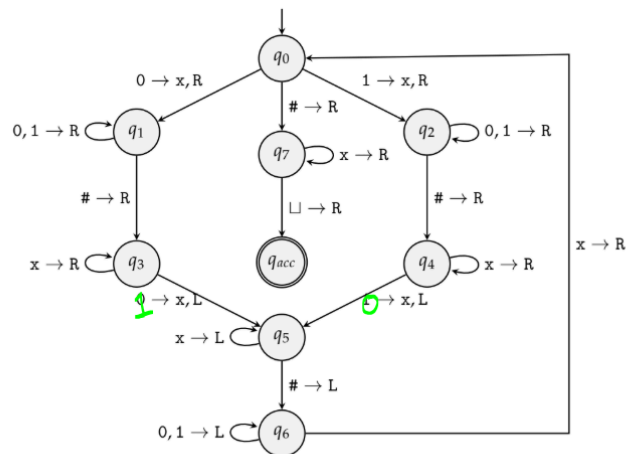
$L_4 = \{ w \in \{0, 1\}^* \mid 0^n 1^{2n} \text{ where } n \geq 0 \}$

$L_5 = \{ w \in \{0, 1\}^* \mid 0^n 1^{2n+2} \text{ where } n \geq 0 \}$

$L_6 = \{ w_1, w_2 \in \{0, 1\}^* \mid w_1 \# w_2 \text{ where } \text{count}(w_1, 0) = \text{count}(w_2, 1) \}$.
 $\text{count}(a, 0)$ means number of 0 in string 'a'.



$L_7 = \{ w_1, w_2 \in \{0, 1\}^* \mid w \# w' \text{ where } w' \text{ is bitwise complement. For eg: if } w \text{ is } 100 \text{ then } w' \text{ is } 011 \}$.



$L_8 = \{ w \in \{0, 1\}^* \mid w \# w^R \# w \}$

3. Pumping Lemma

$$L_1 = \{0^i 1^j 2^k \text{ where } i+j = k\}$$

$$\begin{aligned} &\text{Let } 0^p 1^p 2^{p+p} \\ &= 0^p 1^p 2^{2p} \end{aligned}$$

Assume that L is regular. Let p be the pumping length for L given by the

pumping lemma.

Since $|xy| < 2$, so y has to be 0. If we pump up 0, for example $xyyz$ then we get $0^{p+1} 1^p 2^p$. This is not part of the given language. Hence, proven to be non regular.

$$L_2 = \{0^n 1^m \text{ where } n > m\}$$

Last proof example of the book

$$L_3 = \{0^n 1^m : \text{where } n = 4q + r_1, m = 4q + r_2, q \geq 0 \text{ and } 0 \leq r_1, r_2 \leq 3\}$$

$$L_4 = \{0^n 1^m \text{ where } n/m \text{ is an integer}\} \text{ [here } n \text{ is divided by } m]$$

Assume that L is regular. Let p be the pumping length for L given by the pumping lemma.

$$\text{Let } 0^{p+1} 1^{p+1}, \text{ so } \frac{n}{m} = \frac{p+1}{p+1} = 1, \text{ hence integer}$$

Since $|xy| \leq p$, so y is 0. If we pump down y, we get xz . Hence number of 1 become more than 0, resulting n/m not be an integer.

So it is no longer part of the language, hence non-regular.

$$L_5 = \{ 0^i 1^j 2^k \text{ where if } i=1 \text{ then } j=k \text{ or else } j \neq k \}$$

Assume that L is regular. Let p be the pumping length for L given by the pumping lemma.

If $i=1$, Let $0^1 1^p 2^p$. Since $|xy| \leq p$, y can be 0 or 1 or 01

Case 1:- If y is 0, if we pump up y , we get $xyyz$.

But in this case 0 becomes $\neq 1$, but still $j=k$.

This means $xyyz$ is not part of language. hence non-regular proof.

Case 2:- If y is 1, if we pump up y , we get $xyyz$.

But in this case $j \neq k$ while $i=1$. so we get $0^1 1^{p+|y|} 2^p$. So $xyyz$ is not part of language, hence non-regular.

Case 3:- If y is 01. If we pump up then we get a pattern where 0 appears after 1. so not part of this language. hence non-regular.

4. Turing Machine Proof

$L1 = \{ \langle D, w \rangle : D \text{ is a DFA that accepts } w \text{ and } w \text{ contains odd number of 0's} \}$

$HALT_{TM} = \{ \langle M, w \rangle : M \text{ is a TM that halts on } w \}$

(a) Prove that L1 is decidable.

Similar to TM note proof, only here w always contain 0 number of 0s

(b) Let L_a and L_b be two decidable languages. Prove that, $L_a \cup L_b$ is decidable.

L_a halts on any w , meaning it accepts any string which belongs to the language and rejects any string that does not belong to the language. It never loops forever.

Same thing with L_b .

Construct M_A which simulates a TM that decides L_a . Do same thing with L_b
Run M_A on w . If M_A accepts, accept. If M_A rejects, run M_B on w .
If M_B accepts, accept. If M_B rejects, reject.

(c) Let L_a and L_b be two decidable languages. Prove that, $L_a \cdot L_b$ is decidable.
 L_a halts on any w , meaning it accepts any string which belongs to the language and rejects any string that does not belong to the language. It never loops forever.

Same thing with L_b .

Construct M_A which simulates a TM that decides L_a . Do same thing with L_b

Consider every possible way of splitting w into two parts:
 $w = xy$

Run MAM_A on x . If M_A rejects, try the next split.
If M_A accepts, run MBM_B on y . If M_B accepts, **accept w** .

(d) Prove that, $HALT_{TM}$ is Turing Recognizable but not Turing Decidable.
Already done in the TM notes.

(e) Prove that, the complement of $HALT_{TM}$ is not Turing Recognizable.
(Use proof by contradiction.)